

# Continuing Research on the Assumptions, Models, & Containment Fidelity of Information Union for Fusion of Dependent Data in Square-Root Space



Alexander Michos & Mentor: Dr. Michael Williams (FPS/KAU)

[Alex.Michos.cw@jhuapl.edu] [Michael.Williams@jhuapl.edu] Johns Hopkins University Applied Physics Lab (JHU APL)

**ASPIRE**  
APL STEM EXPERIENCE

**Background:** Track-to-Track-Fusion *combines* the states and covariances (ie ellipsoids) from multiple trackers into a single, more confident estimate. Complex, *interconnected* sensing architectures may introduce *dependencies* between trackers which can violate the assumptions of ‘standard’ fusion schemes, eg the *Square-Root-Information-Filter (SRIF)*, drastically reducing performance.

**Prior Work Summary:** Developed the geometrical and mathematical foundation for the *GEM (Generalized Eigenvalue Method)* algorithm: finding the largest-possible intersection, ie ‘ $\cap$ ’, contained by 2 others. This molded a theory of *Information Union*, ie ‘ $\cup$ ’, by which tracks are fused according to their set-theoretic relationships, even under high dependence/similarity.

**Statement A: Approved for Public Release: Distribution is Unlimited**

Thank you Dr. Williams for your endless support!

## Project Summary

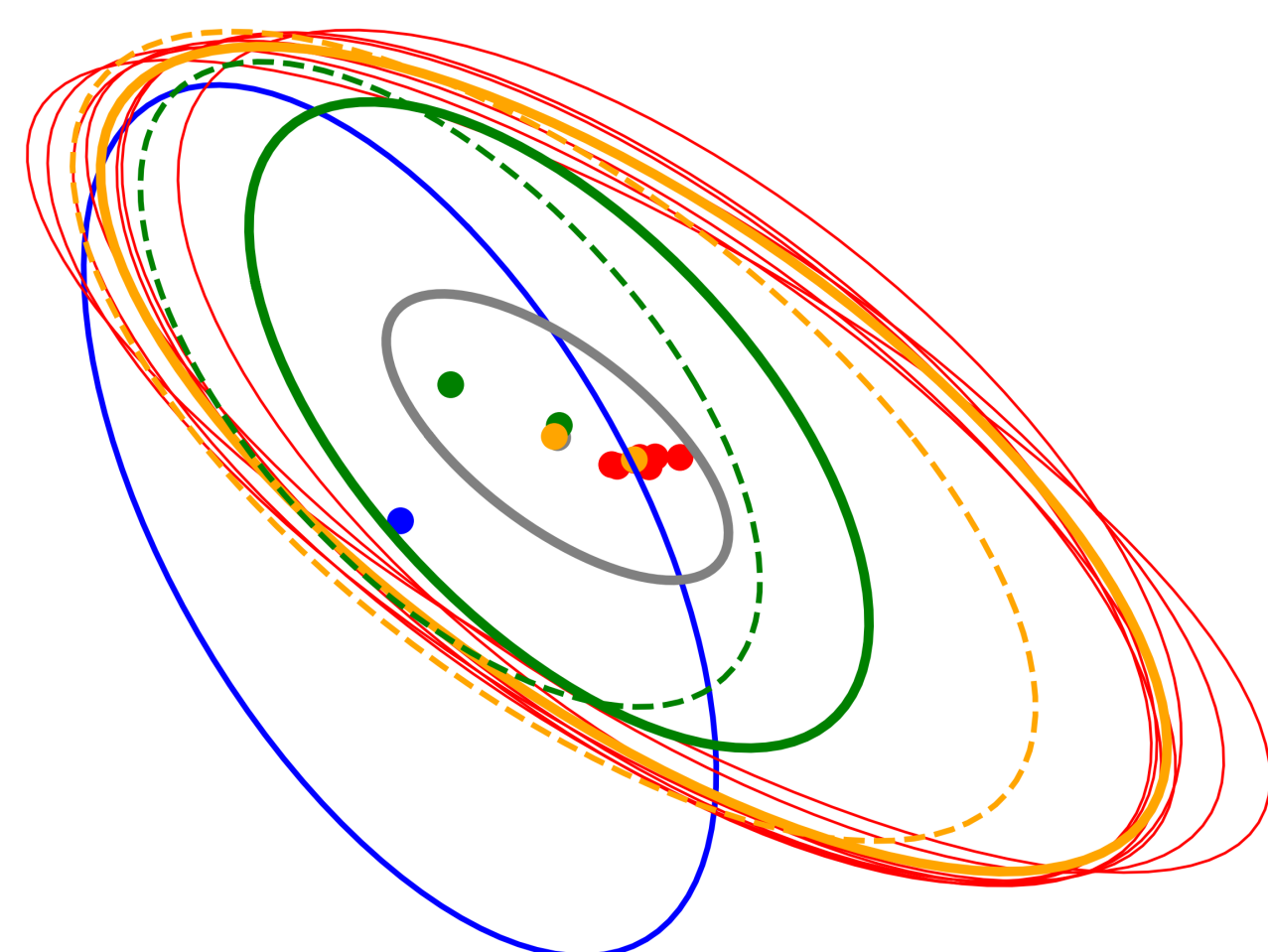
The following lenses explore how statistical *assumptions* on the *formation* of dependencies motivated the construction of mathematical methods for fusion. Each method *extends the theory and applications* of GEM and Info. Union studied in ASPIRE 2024.

**Containment Fidelity** quantitatively refers to how fusers can match *independent* trackers at containing a target with the desired *containment probability*.

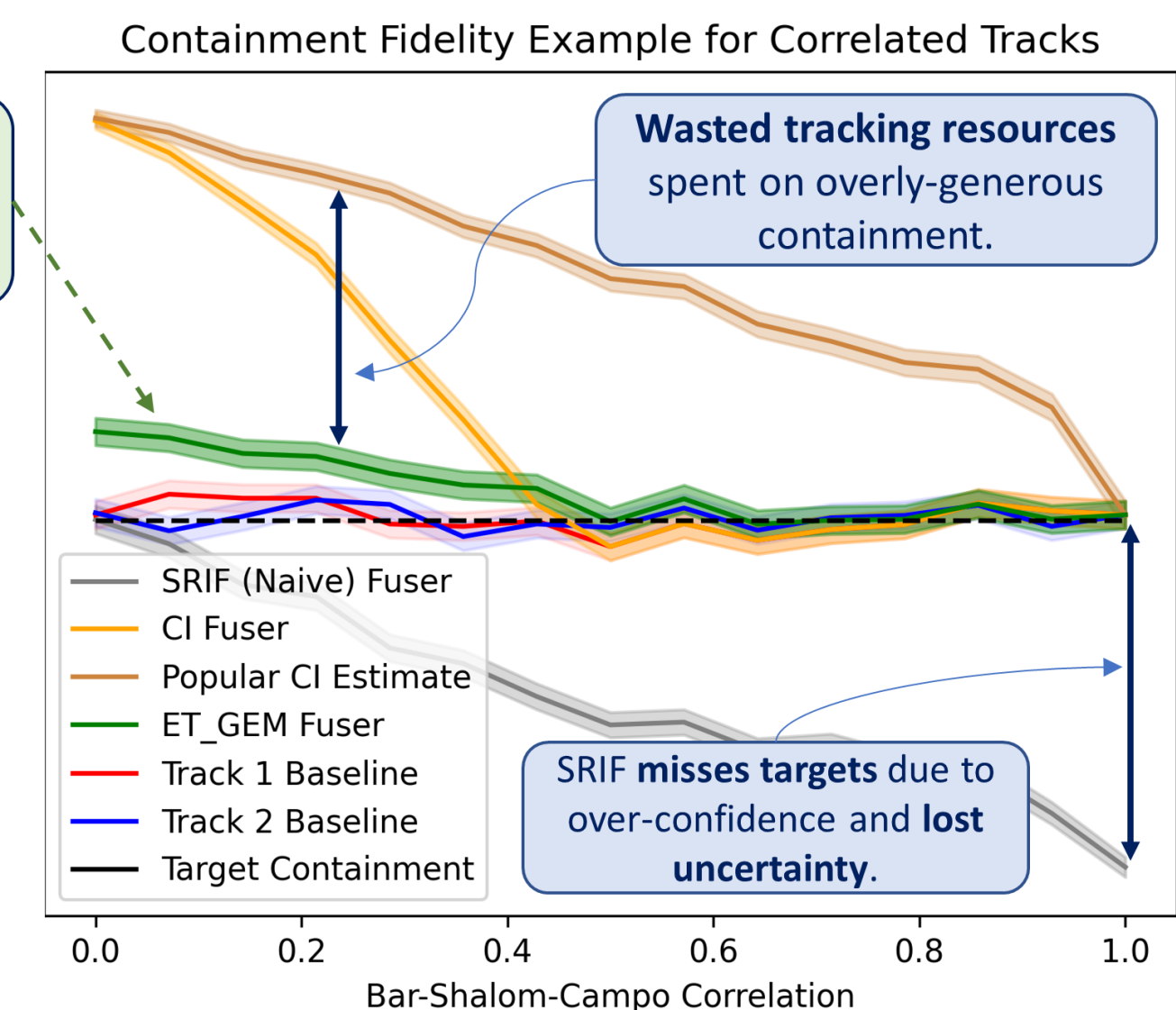
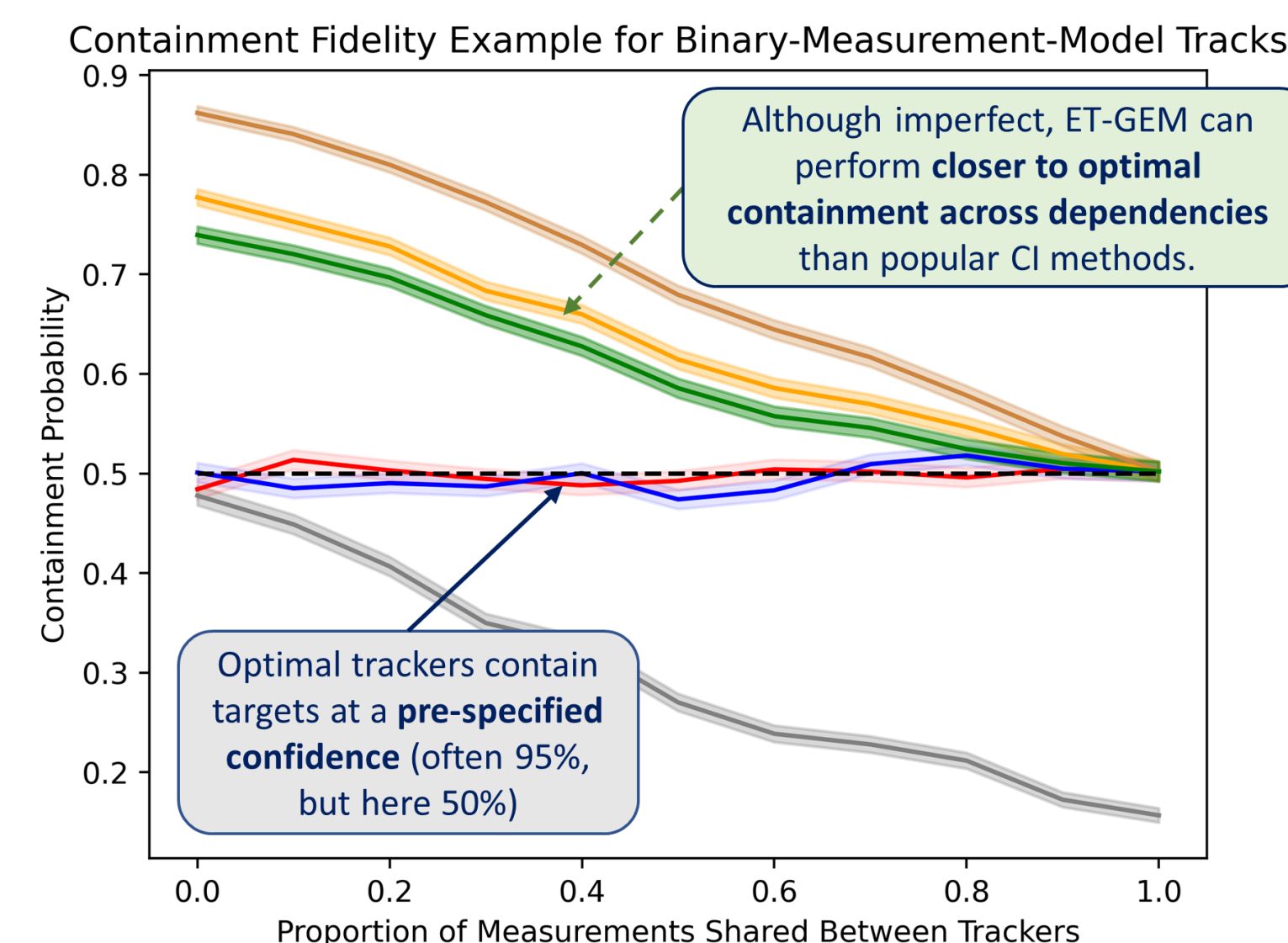
**Open Question:** What is the statistical effect of sequential, pair-wise fusions vs. a joint, global fusion?

$$\text{Fuse}(A_1, \text{Fuse}(A_2, \dots)) \neq \text{Fuse}(A_1, A_2, \dots)$$

See solid (sequential) vs. dashed (joint) ellipses.



Example: Fusion of blue and red tracks using SRIF (grey), GEM (green), and CI (orange).



## Lens 1: Equal Temperament Non-Concentric Fusion

**Assumption:** *Binary Measurement Model:* Measurements are either sourced independently or are duplicated exactly.

**Implication:** Duplications between  $k$  tracks can be estimated by  $\cap$  and excluded with weight  $1/k$ :

$$\bar{I}_i = I_i - \frac{1}{2} \sum_{k \neq i} I_i \cap I_k + \frac{1}{3} \sum_{k_1 < k_2 \neq i} I_i \cap I_{k_1} \cap I_{k_2} - \frac{1}{4} \sum_{k_1 < k_2 < k_3 \neq i} \dots$$

Where the  $\bar{I}_i$ 's evenly *partition* the information union under the Principle of Inclusion-Exclusion:

$$\sum_i \bar{I}_i = \bigcup_i \bar{I}_i = \bigcup_i I_i$$

Allowing fusion of non-concentric means:

$$x_f = (I_f)^{-1} \left( \sum_i \bar{I}_i x_i \right), I_f = \sum_i \bar{I}_i$$

- Previously,  $\bar{I}_i$ 's optimized Likelihood metrics, but this produced undesirable side-effects.
- What is the asymptotic behavior for large  $k$ ? What simplifications can be made?

**Takeaway:** Under the duplication assumption, differing state vectors can be readily fused.

## Lens 2: Fast Multi-Track Union

**Assumption:** *Non-Pairwise Fusion:* Information dependencies in a network cannot be modeled pair-wise.

**Implication:** Need for scalable methods for info. union of many tracks. The border of the interior region to a set of ellipses can be efficiently sampled:

$$v_i = \frac{\tilde{v}_i}{\max_k |R_k \tilde{v}_i|}, \tilde{v}_i \in B(0, 1) \subset \mathbb{R}^N$$

An (un-noramlized) estimate to the largest ellipse contained within the interior (ie GEM) is:

$$\tilde{R}_f = \mathbb{E}(vv^T)$$

More costly is the Linear Program:

$$I_f = \arg \min_I \max_i |v_i^T I v_i - 1| \text{ s.t. } v_i^T I v_i \geq 1 \forall i$$

- Sampling methods scale cheaply with  $k$ , unlike sequential SVD's or QR's.
- How many vectors should be sampled and with what distribution?

**Takeaway:** The largest ellipse within the interior can be rapidly estimated for any  $\#$  of tracks.

## Lens 3: Root Bar-Shalom-Campo

**Assumption:** *Jointly Normal:* Tracks are correlated normal processes rather than corrupted or duplicated.

**Implication:** There exists a set of cross-covariances between state vectors which can be modeled jointly:

$$\mathbf{P} = \begin{bmatrix} P_1 & P_{12} & \dots & P_{1n} \\ P_{12}^T & P_2 & \dots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ P_{1n}^T & \dots & \dots & P_n \end{bmatrix} \Rightarrow \mathbf{R}^T \mathbf{R} = \mathbf{P}^{-1}$$

Using Bar-Shalom-Campo theory, this can be associated with an ‘exact’ square-root fusion. For example,

$$R_f = \text{qr} \begin{bmatrix} R_1 + R_{12} + \dots R_{1n} \\ R_2 + \dots R_{2n} \\ \vdots \\ R_n \end{bmatrix} = \text{qr} \left\{ \rho^{-1/2} \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{bmatrix} \right\}$$

- The SRIF, CI, ICI, GEM family can be reduced to BSC where  $P_1 \sim P_2 \sim P_{12}$  share the GEM basis.
- How can an unknown correlation be estimated?
- *Process Noise* could be a form of correlation, and unlike cross-covariance, is applied exactly.

**Takeaway:** Any model for the correlation of tracks can be directly converted into their fusion.

## Lens 4: Universal Basis Filtering

**Assumption:** *Shared Basis:* Possibly dependent info. matrices have a GEM basis which encodes their commonalities.

**Implication:** Expressing info. matrices in the form  $I_i = J^T D_i J$  gives a class of GEM-like fusion schemes dependent on the basis  $J \in \mathbb{R}^{p \times n}$ .

$$R_f = \text{qr} \left\{ \max \left( \sqrt{D_1}, \dots, \sqrt{D_k} \right) J \right\}$$

Fusion is permutation invariant & contained within the interior region ie  $R_f^T R_f - R_i^T R_i \geq 0 \forall i$ . Given  $k$  tracks, what is the minimum length of this basis?

$$\text{Conjecture: } k \frac{n+1}{2} \approx \min \dim(J) \leq \frac{n(n+1)}{2}$$

- Can *interpolate* between SRIF, CI, Union, etc. by tuning the entries in  $D_f : I_f = J^T D_f J$ .
- $p = \frac{n(n+1)}{2}$  cases give an *associative* fusion.
- Computing  $J$  is related to the Rank- $p$  PARAFAC Tensor Decomposition, where  $\mathbf{T}_{i,:} = I_i$ ,  $\mathbf{T} = D \otimes J^T \otimes J^T$ .

**Takeaway:** Computing a shared basis for tracks extends the desired structure/properties of GEM.